

Design Aspects Of Low-Power Energy Harvesting Circuits

Energy harvesting devices such as piezoelectric elements and antennas capture ambient energy, but this energy has to be converted to DC for load and energy storage devices. For this high-efficiency power harvesting circuits are required. This article discusses important design aspects of energy harvesting circuits and their implementation

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Recent developments in ultra-low-power circuit designs have reduced the power consumption of portable electronic devices and low-power wireless sensors significantly. The power now required has been reduced to just picowatts in the standby mode and nanowatts in the active mode, with the devices operating on an ultra-low duty cycle. These breakthroughs harvest energy from ambient vibrations, radio frequency, static electricity, and other ambient sources, making the best use of energy harvesting solutions.

The energy harvesters make it possible to achieve battery-less operation in low-power devices. The previous article on battery-less solutions discussed some popular energy harvesting techniques that produce enough power to run specific low-power devices. But after capturing the power from source, it needs to be rectified and conditioned before it is fit for use.

A typical ambient energy harvesting circuit has a transducer that translates the ambient energy into electrical energy. In the case of vibrational energy harvesters, these transducers are piezoelectric elements, mechanical oscillators, electromagnetic vibrational

harvesters, etc. In the case of RF energy harvesting, the transducer is the antenna. Likewise, there are various other transducers for different energy sources.

The energy converted by the transducers is rectified to generate a DC voltage suitable for load and supercapacitors (for energy storage). The main aim while designing energy harvesting circuits is to maintain high efficiency at every stage. To ensure that, we need an impedance matching circuit between the transducer and the rectifier for maximum power transfer.

After rectification, the voltage may be multiplied by voltage multiplier circuits for compatibility. Now, most of the battery-less low-power devices employ supercapacitors for energy storage. To facilitate such storage, a switch-mode DC-DC converter circuit can be used. Some sophisticated systems use adaptive energy storage systems that compensate for the shortcomings of single energy storage devices.

For ensuring smooth output and high efficiency, feedback circuits can be used to manage the converter circuits. The goal is not only to optimise the power harvested but also to maximise the energy conversion efficiency.

With this, we start discussing the very first stage that ensures maximum power transfer from the transducer to the rectifier circuit—an impedance matching circuit.

Impedance matching

Impedance matching circuit is crucial in optimising the performance of the energy harvesting system. Mismatch in the impedance of transducers and rectifiers degrades the power conversion efficiency of the energy harvester circuit.

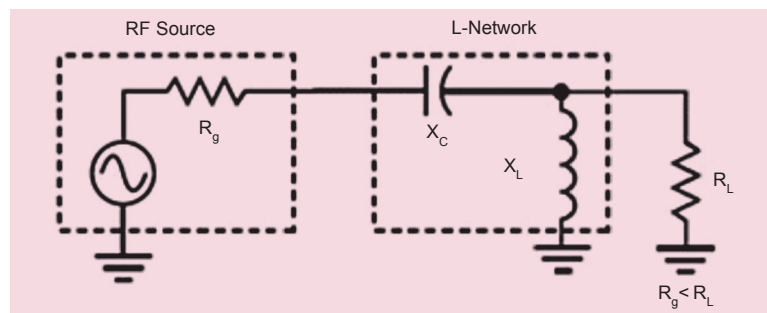


Fig. 1: LC based impedance matching